

$$L_{\rm SAT} = \sum_{i,o} \left[\frac{1}{2} \mu_i \left| \frac{d\lambda}{dH} \right|^2 + \sum_j \kappa F_{\rm braid} + \alpha \left(\frac{d\lambda}{dH} \cdot T \right)^2 + V_{\rm geom}(\lambda) \right]$$

$$L_{\rm UI} = \frac{1}{2}(\partial_\lambda r)^2 + \frac{1}{2}r^2\Omega_{\mu\nu}\Omega^{\mu\nu}$$

$$\Omega^\mu{}_\nu = (\partial_\lambda R^\mu{}_\alpha) R^\alpha{}_\nu, \quad \Omega_\mu = A_\mu^a T^a + B_\mu^i \tau^i + C_\mu Y$$

$$F_{\mu\nu} = \partial_\mu \Omega_\nu - \partial_\nu \Omega_\mu + [\Omega_\mu, \Omega_\nu], \quad J_\mu = *J_{\mu\nu\rho}$$

$$L_u = \int \left[\frac{1}{2} (\partial u)^2 + U(u) \right] d^4x, \quad K = \kappa_0 + \kappa_2$$